Chapter 4: Introduction to R

R for Descriptive Statistics

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0.1 Overview

This reading prepares you for the lab session where you'll use R to compute descriptive statistics. You don't need any prior programming experience — we'll use WebR, which runs directly in your browser.

0.2 What Is R?

R is a programming language designed for statistical analysis. It's:

- **Free and open-source** — anyone can use it
- **Powerful** — handles everything from simple means to complex models
- **Widely used** — the standard in psychology, biology, medicine, and data science
- **Reproducible** — your analysis is documented in code

0.2.a Why Learn R?

Excel can calculate a mean. So why bother with R?

1. **Reproducibility**: R code documents exactly what you did. "I calculated the mean" becomes `mean(data$reaction_time)` — clear and precise.
2. **Efficiency**: Once written, code can be reused. Analyse one dataset, apply the same analysis to another.
3. **Capabilities**: R handles analyses that Excel simply cannot — bootstrapping, mixed models, complex visualisations.
4. **Transparency**: Sharing code lets others check your work. This is increasingly required in psychology journals.

0.3 WebR: R in Your Browser

We use **WebR** — R that runs entirely in your browser. No installation needed.

0.3.a How It Works

When you see a code box like this:

```
{webr-r}
mean(c(5, 10, 15, 20))
```

You can:

1. Click in the box
2. Press the "Run" button
3. See the output immediately

The code runs in your browser, not on a server. Your data stays on your machine.

0.4 R Basics for Today

You don't need to master R today — just understand enough to follow along. Here are the essentials:

0.4.a Assigning Values

Use `<-` (less-than followed by dash) to assign values:

```
{webr-r}
# Store a value
my_score <- 75

# Store multiple values
reaction_times <- c(245, 312, 278, 298, 267)
```

The `c()` function combines values into a **vector** (a list of numbers).

0.4.b Basic Functions

Functions do things. They have a name followed by parentheses containing inputs:

```
{webr-r}
# Mean of values
mean(c(10, 20, 30))

# Sum of values
sum(c(10, 20, 30))

# Number of values
length(c(10, 20, 30))
```

0.4.c Computing Descriptives

For descriptive statistics, you'll use:

Function	What It Does
`mean(x)`	Arithmetic mean
`median(x)`	Median (middle value)
`sd(x)`	Standard deviation

Function	What It Does
`var(x)`	Variance
`min(x)`	Minimum value
`max(x)`	Maximum value
`range(x)`	Returns both min and max
`summary(x)`	Overview (min, quartiles, mean, max)

0.4.d Example: Reaction Times

```
{webr-r}
# Create data
rt <- c(245, 312, 278, 298, 267)

# Compute statistics
mean(rt)     # 280
median(rt)   # 278
sd(rt)       # 26.2
```

0.5 Understanding Output

When R gives you output, it includes some formatting. For example:

```
[1] 280
```

The `[1]` just means "first element." For a single number, you can ignore it.

For longer output:

```
[1] 245 267 278 298 312
```

The numbers 245, 267, etc. are your data. The `[1]` tells you where the first element is.

0.6 Working with Data Frames

Real datasets aren't just vectors — they're **data frames** (like a spreadsheet with rows and columns).

0.6.a Accessing Columns

Use `$` to access a column:

```
{webr-r}
# Create example dataset
olympics <- data.frame(
  reaction_time = c(245, 312, 278, 298, 267),
  room = c("A", "A", "B", "B", "A")
```

```
)

# Get the reaction_time column
olympics$reaction_time

# Mean of that column
mean(olympics$reaction_time)
```

0.6.b Subsetting Data

Get specific parts of your data:

```
{webr-r}
# Using the olympics data from above
# Filter rows where room is "A"
room_a <- olympics[olympics$room == "A", ]

# Get mean reaction time for Room A only
mean(room_a$reaction_time)
```

0.7 Common Errors (And How to Fix Them)

0.7.a Error: object 'x' not found

Meaning: R doesn't recognise that name.

Fixes:

- Check spelling (R is case-sensitive: `Data` ≠ `data`)
- Make sure you created the object first
- Run all code chunks in order

0.7.b Error: could not find function "x"

Meaning: R doesn't recognise that function.

Fixes:

- Check spelling
- Some functions need packages (we'll load these for you)

0.7.c Unexpected symbol error

Meaning: R found something it didn't expect.

Fixes:

- Check for missing commas or parentheses
- Make sure quotes match (both `"` or both `'`)

0.8 Preparing for the Lab

0.8.a What You'll Do

In the lab, you'll:

1. Load the DataLab dataset
2. Explore the variables (columns)
3. Calculate means, medians, and SDs for different measures
4. Compare Room A vs Room B
5. Visualise distributions with histograms

0.8.b Before You Arrive

- Read this page (you're doing that now!)
- Have the Lecture Reading fresh in mind
- Bring a laptop or phone (WebR runs in any modern browser)

0.8.c What to Focus On

The goal isn't to memorise R syntax. Focus on:

- **What** each statistic tells you (meaning over mechanics)
- **How** the statistics relate to your DataLab experience
- **Why** we compute these particular measures

0.9 Key Terms

Term	Definition
R	Programming language for statistical analysis
WebR	R running in a web browser
Vector	A list of values of the same type
Data frame	A table with rows and columns
Function	Code that performs an action (like `mean()`)

0.10 Summary

What	How in R
Mean	`mean(x)`
Median	`median(x)`
SD	`sd(x)`
Access column	`data$column`
Create vector	`c(1, 2, 3)`

0.11 Additional Resources

If you want to explore R further:

- R for Data Science — Free online textbook
- RStudio Primers — Interactive tutorials
- Quick-R — Syntax reference